

Mohak Gupta

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PROFILE SUMMARY

Full-stack engineer who builds **AI-powered systems end-to-end** — from prompt architecture and LLM pipelines to shipping and monitoring in production. I use **Cursor, Claude Code, and GitHub Copilot** in every session as development infrastructure, not as occasional helpers. Won **InnovCon Hacknex 2025** and placed runner-up at **Smart India Hackathon 2023** shipping real products under pressure.

SKILLS

Languages: TypeScript, Python, Go, C++, Java, Rust, Solidity

AI & LLMs: LLM integration, prompt chaining, RAG pipelines, structured output extraction, context management

Backend: Bun.js, Express, Gin; async workflows, REST APIs, JWT/RBAC, WebSockets

Frontend: React, Next.js

Cloud & DevOps: GCP, Docker, Kubernetes, Pub/Sub, RabbitMQ

Databases: PostgreSQL, MongoDB, Redis

EDUCATION

Techno India University, Kolkata

B.Tech in Computer Science (AI Minor)

July 2023 – Present

CGPA: 8.88

EXPERIENCE

Software Engineer | Valstrand

Dec 2025 – Present

- o Built **CallGenie**, an AI voice calling agent handling automated inbound and outbound conversations for real estate workflows — designed the prompt architecture, managed context across long sessions, and owned the system through to production monitoring.
- o Developed **Airmeet AI Agent**, a domain-specific voice agent grounded in Airmeet data; implemented retrieval logic to keep responses accurate and on-topic.
- o Shipped **RealtyCRM** (lead tracking and pipeline management) and **RealtyBuilder** (multi-tenant landing page builder) as full-stack features — wrote the code, deployed with Docker, and debugged what broke in production.
- o Designed and built **realtor-focused website templates** with embedded voice agents and MLS/IDX integration for Canadian markets.

Cybersecurity & Blockchain Lead | GDG On Campus

June 2024 – Nov 2025

- o Ran workshops on hacking fundamentals, reconnaissance, and smart contract security; mentored students through hackathon projects end-to-end.

PROJECTS

ChainWork — Decentralized Freelance Marketplace

Winner, InnovCon Hacknex 2025

- o Built a trustless freelance platform using **Solidity smart contracts** with escrow-based payments and on-chain milestone validation — no central authority, no intermediary.
- o Implemented **real-time communication** via WebSockets, REST APIs with **JWT auth and RBAC** for contract and user management. Used AI coding tools to accelerate scaffolding; took full responsibility for debugging and production behavior.
- o **Stack:** React, Express, Solidity, Hardhat, Socket.io

RAG Pipeline — Retrieval Augmented Generation from Scratch

(WIP)

- o Building a RAG system from the ground up — document ingestion, tokenization, inverted indexing, vector search, and query processing — to understand exactly what happens before the LLM sees context.
- o Designed evaluation criteria for output quality: not just “does it answer the question” but “is the retrieved context actually relevant and is the generation grounded in it.”
- o **Stack:** Python

LaunchT — Solana Token Launchpad

- o Wallet-integrated launchpad on Solana Devnet for creating and managing **SPL tokens**; handled async signing flows and on-chain transactions using **@solana/web3.js**.
- o **Stack:** TypeScript, Next.js, @solana/web3.js

CERTIFICATIONS

- o **CompTIA PenTest+** — Vulnerability scanning, exploitation techniques, and penetration testing across network and application environments. **Mar 2023**
- o **Certified Network Security Professional (CNSP)** — Network defense, encryption standards, and firewall configuration. **Mar 2023**